THE SHANNON–WAKIL EFFECT: CONSTITUTIONAL FORCING AS A UNIVERSAL PATTERN IN INFORMATION THEORY AND PRIME ARITHMETIC

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Abstract. We formalize a structural pattern that we call the *Shannon–Wakil effect*: an exponential configuration space on which a natural measure concentrates onto a strict subspace whose size, on the exponential scale, is a fixed fraction $\theta^* \in (0,1)$ of the whole. Shannon's Asymptotic Equipartition Property (AEP) is an instance: for a source with $0 < H < \log_2 |\mathcal{A}|$, the $|\mathcal{A}|^n$ sequences concentrate onto a typical set of size $2^{n(H+o(1))}$, so $\theta^* = H/\log_2 |\mathcal{A}|$, although the typical set is a vanishing fraction of $\mathcal{A}^n$. Shannon's Channel Coding Theorem shares only the size condition: the largest reliable codebook in $\mathcal{X}^n$ has $2^{n(C+o(1))}$ codewords, with $C = \max_{p(x)} I(X;Y)$, but no measure given by the channel concentrates on it. These constants vary continuously with the source and the channel and are not values of the k-hierarchy $\theta_k = (2^k - k)/2^k$.

The first version also claimed a third, arithmetic instance: primes modulo powers of 3, governed by $\theta_W = 5/8$. That claim is withdrawn. Primes are equidistributed between the residue classes 1 and 2 modulo 3, so they do not concentrate on the proposed subspace; the summed estimate over the moduli 3^K holds for every exponent up to 1; and $\theta_W = 5/8$ for these moduli is a conjecture of the companion paper [12]. The Sophie Germain "non-instance" cannot discriminate between depths, since $\theta_1 = \theta_2 = 1/2$, and is withdrawn as a test. We record the remaining parallels as informal analogies and state the Constitutional Forcing Conjecture, in which the counting formula is a postulate.

Revision note (November 2026). The arithmetic "third instance" is withdrawn: primes do not concentrate on the class 2 mod 3 (Proposition 5.2 now shows that (SW3) fails), Theorem 5.1 of the first version is replaced by the trivial estimate that it is (Proposition 5.1), and $\theta_W = 5/8$ is stated as a conjecture. The Sophie Germain "theorem", its citation of itself and the unreproducible statistic $\delta \cdot \log x = 2.38 \pm 0.18$ are withdrawn, and so is the claim that information theory supplies $k = 1$ instances. The false capacity formula $C = \log |\mathcal{X}| - \min H(X|Y)$ is replaced by $C = \max_{p(x)} I(X;Y)$, and the degraded-channel remark is corrected. Proposition 3.2 now carries the hypotheses it needs, and channel coding is recast on $\mathcal{X}^n$, where it satisfies only the size conditions. The "five structural correspondences" and the "[Proven]" labels on them are replaced by informal analogies, and the "three-level filtration", the "8 configuration dimensions" and the 5/4 "gain" are removed. The Bombieri–Vinogradov and Pascadi references and a duplicate label are corrected.

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1. Introduction

Every mathematician knows at some level that information theory and number theory are different subjects. Information theory studies the storage and transmission of data; number theory studies the arithmetic structure of integers. They have different objects, different tools, and different communities.

This paper proposes a structural parallel between the two and tests it.

In 1948 Shannon published the two results that define the fundamental limits of compression and communication [1]. The Asymptotic Equipartition Property (AEP) says that most sequences are typical: the $|\mathcal{A}|^n$ possible sequences of length n concentrate on an effective set of size $2^{n(H+o(1))}$, governed by the entropy H; when $H < \log_2 |\mathcal{A}|$ this set is a vanishing fraction of $\mathcal{A}^n$. The Channel Coding Theorem says that reliable communication at rate R is possible for $R < C$ and impossible for $R > C$, where C is the channel capacity. Shannon's 1948 argument for the coding theorem was a random-coding sketch; rigorous proofs followed (Feinstein, 1954). In both cases the governing constant is a continuous function of the source or channel statistics.

The first version of this paper proposed an analogous arithmetic instance: that the algebraic structure of $\mathbb{Z}[\omega]$ forces primes in arithmetic progressions modulo the cascade moduli 3^K to concentrate onto an effective subspace, governed by $\theta_W = 5/8$. Section 5 shows that this proposal fails: primes are equidistributed between the classes 1 and 2 modulo 3, the summed estimate behind it holds for every exponent up to 1, and $\theta_W = 5/8$ for these moduli is a conjecture [12].

1.1. **Two instances and a withdrawn third.**

(1) **Shannon AEP (1948).** For $0 < H < \log_2 |\mathcal{A}|$, source sequences of length n concentrate onto $2^{n(H+o(1))}$ typical sequences out of $|\mathcal{A}|^n$. Governing constant: $H/\log_2 |\mathcal{A}|$, a continuous function of the source distribution p (Proposition 3.2).

(2) **Shannon Channel Coding (1948).** Reliable codes exist at all rates $R < C$ and not at rates $R > C$, where $C = \max_{p(x)} I(X;Y)$. Governing constant: $C/\log_2 |\mathcal{X}|$, a continuous function of the channel $p(y|x)$. This case satisfies only the size conditions of the definition (Proposition 4.2).

(3) **Cascade moduli (proposed in the first version).** This is not an instance (Proposition 5.2); $\theta_W = 5/8$ for the moduli 3^K is conjectural [12].

Sophie Germain primes, presented in the first version as a confirmed non-instance, are discussed in Section 7: the example cannot discriminate between depths.

1.2. **What this paper claims and does not claim.** We attach logical-status labels to every significant claim. Here we state the top-level ones.

[Proven] Under the hypotheses $0 < H < \log_2 |\mathcal{A}|$ and $\varepsilon_n \to 0$ with $n\varepsilon_n^2 \to \infty$, the AEP satisfies (SW1)–(SW4) with $\theta^* = H/\log_2 |\mathcal{A}|$ (Proposition 3.2).

[Proven] Channel coding, recast on $\mathcal{X}^n$, satisfies the size conditions with $\theta^* = C/\log_2 |\mathcal{X}|$ but not the concentration condition (Proposition 4.2).

[Proven] The cascade-moduli system does not satisfy (SW3) (Proposition 5.2), and the summed estimate of the first version holds for every exponent up to 1 (Proposition 5.1).

[Conjectured] $\theta_W = 5/8$ for the moduli 3^K, per modulus [12]; this would go beyond the Generalised Riemann Hypothesis.

[Heuristic] The Sophie Germain remark of Section 7; it is not a test of the framework.

[Analogy] The parallels of Section 6. They are informal analogies, not theorems.

[Conjectured] The Constitutional Forcing Conjecture (Conjecture 9.1), whose hypothesis (iii) is a postulate.

[Open] Specifying any object for $k = 4$ (Remark 11.1); without one, $\theta_4 = 3/4$ is not a testable prediction.

[Not claimed] A functorial or categorical equivalence between information theory and number theory.

1.3. **Position in the programme.** This paper is the sixth in a nine-paper programme (p01–p09) on constitutional sieve theory [9, 10, 11, 12, 13, 14]. The formula $\theta_k = (2^k - k)/2^k$ is obtained in [9, 13] from a counting postulate. The level $\theta_W = 5/8$ for the moduli 3^K is a conjecture of [12]. Condition W3 of [14] is true, with a one-line proof from the cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^*$, and does not bear on any level of distribution. The methodological framework is articulated in [11].

2. The Shannon–Wakil Effect: Definition

Definition 2.1 (Shannon–Wakil Effect)**.** A system $\mathcal{S} = (\mathcal{C}_n, \mu_n)_{n\geq 1}$, consisting of finite sets $\mathcal{C}_n$ and probability measures μ_n on them, exhibits the *Shannon–Wakil effect* with governing constant $\Phi(\mathcal{S}) = \theta^*$ if:

(SW1) **Exponential configuration space.** $|\mathcal{C}_n|$ grows exponentially in n: $|\mathcal{C}_n| = e^{n(r+o(1))}$ for some $r > 0$.

(SW2) **Intrinsic structural constraint.** μ_n is a measure or distributional law intrinsic to $\mathcal{S}$—not an external parameter.

(SW3) **Dimensional reduction.** There exist effective subspaces $\mathcal{E}_n \subsetneq \mathcal{C}_n$ with $|\mathcal{E}_n| = e^{n(\theta^* r + o(1))}$ for some $\theta^* \in (0,1)$, onto which μ_n concentrates: $\mu_n(\mathcal{C}_n \setminus \mathcal{E}_n) \to 0$.

(SW4) **Determination of the governing constant.** The constant θ^* is determined by the data of $\mathcal{S}$.

Remark 2.2. (SW1) and (SW3) are the mathematical content of Definition 2.1. (SW2) and (SW4) are informal requirements: "intrinsic" is not defined, and any quantity defined from $\mathcal{S}$ is determined by $\mathcal{S}$, so (SW4) adds nothing beyond requiring θ^* to be defined from the system. The first version called (SW4) "the critical one", required θ^* to be "uniquely determined by the algebraic structure", and asserted that in the cascade case changing θ^* "would require changing the prime 3 itself". Entropy and capacity are continuous functions of real parameters, not algebraic invariants, and the cascade case is withdrawn (Section 5); those statements are withdrawn.

3. First Instance: Shannon's AEP

Let $\mathcal{A}$ be a finite alphabet with probability distribution p on $\mathcal{A}$. Let $X_1, X_2, \ldots$ be i.i.d. with law p. The entropy is $H = H(p) = -\sum_{a\in\mathcal{A}} p(a)\log_2 p(a)$.

Theorem 3.1 (AEP, [2, Theorem 3.1.2])**.** *For any $\varepsilon > 0$, the typical set*

$$T_\varepsilon^{(n)} = \left\{ x^n \in \mathcal{A}^n : \left| -\frac{1}{n}\log_2 p(x^n) - H \right| < \varepsilon \right\}$$

satisfies: (i) $\Pr[X^n \in T_\varepsilon^{(n)}] \to 1$; (ii) $|T_\varepsilon^{(n)}| \le 2^{n(H+\varepsilon)}$; (iii) *for large n,* $|T_\varepsilon^{(n)}| \ge (1-\varepsilon)2^{n(H-\varepsilon)}$.

Proposition 3.2 (AEP Exhibits SW Effect)**.** *Suppose $0 < H(p) < \log_2|\mathcal{A}|$, and let $\varepsilon_n \to 0$ with $n\varepsilon_n^2 \to \infty$. Then the system $\mathcal{S}_{AEP} = (\mathcal{A}^n, p^{\otimes n})$ satisfies* (SW1)–(SW4) *with effective subspaces $\mathcal{E}_n = T_{\varepsilon_n}^{(n)}$ and governing constant $\theta^* = H/\log_2|\mathcal{A}|$.*

Proof. (SW1): $|\mathcal{A}^n| = e^{nr}$ with $r = \ln|\mathcal{A}| > 0$. (SW2): $p^{\otimes n}$ is the i.i.d. law of the source. (SW3): $-\frac{1}{n}\log_2 p(X^n)$ is the mean of n i.i.d. copies of $-\log_2 p(X_1)$, which has mean H and finite variance σ^2, so by Chebyshev's inequality $\Pr[X^n \notin T_{\varepsilon_n}^{(n)}] \le \sigma^2/(n\varepsilon_n^2) \to 0$. Every $x^n \in T_{\varepsilon_n}^{(n)}$ has $2^{-n(H+\varepsilon_n)} < p(x^n) < 2^{-n(H-\varepsilon_n)}$, so $(1-o(1))\,2^{n(H-\varepsilon_n)} \le |T_{\varepsilon_n}^{(n)}| \le 2^{n(H+\varepsilon_n)}$. Hence $|T_{\varepsilon_n}^{(n)}| = 2^{n(H+o(1))} = e^{n(\theta^* r + o(1))}$ with $\theta^* = H/\log_2|\mathcal{A}| \in (0,1)$, and $T_{\varepsilon_n}^{(n)} \neq \mathcal{A}^n$ once $\varepsilon_n < \log_2|\mathcal{A}| - H$. (SW4): θ^* is a function of p. □

Remark 3.3. The hypotheses are needed. For the uniform source every x^n has $-\frac{1}{n}\log_2 p(x^n) = H$, so the typical set is all of $\mathcal{A}^n$ and the exponent ratio is 1; for a deterministic source it is 0. With ε fixed, Theorem 3.1 pins $|T_\varepsilon^{(n)}|$ only between $2^{n(H\pm\varepsilon)}$. The typical set is a vanishing fraction of $\mathcal{A}^n$: $|T_{\varepsilon_n}^{(n)}|/|\mathcal{A}|^n = 2^{-n(\log_2|\mathcal{A}| - H + o(1))} \to 0$. The governing constant $H/\log_2|\mathcal{A}|$ takes every value in $(0,1)$ as p varies (for example $0.918\ldots$ for $p = (1/3, 2/3)$); it is not $1/2$ in general.

4. Shannon's Channel Coding Theorem: A Partial Instance

A discrete memoryless channel is a conditional distribution $p(y|x)$ from input alphabet $\mathcal{X}$ to output alphabet $\mathcal{Y}$. The channel capacity is

$$C = \max_{p(x)} I(X;Y) = \max_{p(x)} \sum_{x,y} p(x)p(y|x)\log\frac{p(y|x)}{p(y)}.$$

Theorem 4.1 (Channel Coding, [2, Ch. 7])**.** (Achievability) *For any $R < C$ and $\varepsilon > 0$, there exist codes of rate R and blocklength n with maximal error probability $< \varepsilon$ for all sufficiently large n.*

(Converse) *For any $R > C$, every sequence of codes of rate R has error probability bounded away from 0.*

The first version took as configuration space the set of all codebooks of rate R and blocklength n. There are $|\mathcal{X}|^{n\cdot 2^{nR}}$ of them, which is doubly exponential in n, so that choice violates (SW1); the first version nevertheless asserted (SW1). It also took the channel law $p(y|x)$ as the measure, but $p(y|x)$ is not a measure on codebooks. We use instead $\mathcal{C}_n = \mathcal{X}^n$, in which a codebook is a subset. For $0 < \varepsilon < 1$ let $M^*(n, \varepsilon)$ be the largest number of codewords in a code of blocklength n with maximal error probability at most ε.

Proposition 4.2 (Channel coding and the SW conditions)**.** *Suppose $0 < C < \log_2 |\mathcal{X}|$. Then*

$$\lim_{\varepsilon\to 0} \liminf_{n\to\infty} \tfrac{1}{n} \log_2 M^*(n,\varepsilon) = \lim_{\varepsilon\to 0} \limsup_{n\to\infty} \tfrac{1}{n} \log_2 M^*(n,\varepsilon) = C.$$

Thus a largest codebook $\mathcal{E}_n \subsetneq \mathcal{X}^n$ with maximal error at most ε has $|\mathcal{E}_n| = 2^{n(C+o(1))}$ in this iterated sense, and (SW1) and the size part of (SW3) hold with $\theta^* = C/\log_2 |\mathcal{X}|$. The concentration part of (SW3) has no counterpart: no measure on $\mathcal{X}^n$ given by the channel concentrates on $\mathcal{E}_n$. Channel coding is therefore not an instance of Definition 2.1 as stated.

Proof. By achievability, $\liminf_n \frac{1}{n} \log_2 M^*(n,\varepsilon) \geq R$ for every $R < C$ and every $\varepsilon > 0$. A code with maximal error at most ε has average error at most ε, so Fano's inequality gives $(1-\varepsilon)\log_2 M \leq nC + 1$ for its size M, and $\limsup_n \frac{1}{n} \log_2 M^*(n,\varepsilon) \leq C/(1-\varepsilon)$. Let $\varepsilon \to 0$. □

Remark 4.3. The AEP satisfies Definition 2.1 (Proposition 3.2); channel coding satisfies only its size conditions (Proposition 4.2). The first version took the fact that one definition "captures both source coding and channel coding limits" as evidence that Definition 2.1 is the right one; that inference is withdrawn. As with the AEP, the constant $C/\log_2 |\mathcal{X}|$ varies continuously with the channel: for the binary symmetric channel with crossover probability ϵ it is $1 - H(\epsilon)$, which equals $1/2$ only at $\epsilon \approx 0.110$.

5. The Proposed Third Instance: Cascade Moduli

The first version stated, as its third instance, the estimate below with the range $3^K \leq x^{5/8-\varepsilon}$, citing Theorem 1.1 of the first version of [12]. It asserted that 5/8 is "uniquely forced" by a "three-level Eisenstein filtration" of $\mathbb{Z}[\omega]/(3^K)$ and the cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^*$, and that no $\theta > 5/8$ is achievable for cascade moduli. In fact the estimate holds for every exponent up to 1.

Proposition 5.1 (The summed estimate is trivial)**.** *For every $A > 0$,*

$$\sum_{\substack{K\geq 1\\ 3^K\leq x}} \max_{\gcd(a,3)=1} \left| \sum_{\substack{n\leq x\\ n\equiv a\,(3^K)}} \Lambda(n) - \frac{x}{\varphi(3^K)} \right| \ll_A \frac{x}{(\log x)^A}.$$

Proof. Write $\psi(x; q, a)$ for the inner sum and let $L = (\log x)^{A+1}$. For $3^K \leq L$ the Siegel–Walfisz theorem gives $\max_a |\psi(x; 3^K, a) - x/\varphi(3^K)| \ll_A x \exp(-c\sqrt{\log x})$, and there are $O(\log\log x)$

such K. For $3^K > L$, the trivial bounds $\psi(x;q,a) \le (x/q+1)\log x$ and $x/\varphi(3^K) = 3x/(2\cdot 3^K)$ give $|\psi(x;3^K,a) - x/\varphi(3^K)| \le (x/3^K+1)\log x + 3x/(2\cdot 3^K)$. Summing over the $O(\log x)$ values of K with $L < 3^K \le x$ gives $\ll x\log x/L + (\log x)^2 \ll x/(\log x)^A$. □

So the estimate of the first version is true, but its exponent plays no role, and the claims that 5/8 is "uniquely forced" and that "no $\theta > 5/8$ is achievable" are withdrawn. The $(1-\omega)$-adic filtration of $\mathbb{Z}[\omega]/(3^K)$ has $2K$ steps, each with quotient $\mathbb{F}_3$, not three levels, and it does not determine the value 5/8. The meaningful statement is per modulus: for $q = 3^K \le x^{5/8-\varepsilon}$, $\max_a |\psi(x;q,a) - x/\varphi(q)| \ll x/(\varphi(q)(\log x)^A)$. This is a conjecture of [12]. For a single modulus the Generalised Riemann Hypothesis gives it only for $q \le x^{1/2-\varepsilon}$, and the best known results for prime-power moduli are those of Barban–Linnik–Tshudakov (exponent 3/8), Gallagher (2/5) and Huxley (5/12); Banks and Shparlinski obtain lower bounds of the correct order for moduli up to $x^{0.4736}$. Pascadi's exponent 5/8 [7] concerns smooth numbers, and primes with triply-well-factorable weights, in arithmetic progressions to general moduli; it is obtained by spectral methods and is unrelated to the moduli 3^K or to $\mathbb{Z}[\omega]$. Its agreement with θ_3 is a numerical coincidence.

In [9, 13] the value $5/8 = (2^3-3)/2^3$ comes from a counting postulate: a system of depth k has exactly k of its 2^k binary configurations eliminated. It is not derived: if "invalid" means "some constraint fails", there are $2^k - 1$ invalid configurations. The assignment of depth 3 to cascade moduli or twin primes is proposed and currently unsupported: for $p > 3$ twin primes satisfy only $p \equiv 2 \pmod 3$, and the conditions modulo 9 and 27 are implied by it.

Proposition 5.2 (Cascade moduli do not exhibit the SW effect)**.** *Let $\mathcal{C}_K = (\mathbb{Z}/3^K\mathbb{Z})^*$, let $\mathcal{E}_K = \{a \in \mathcal{C}_K : a \equiv 2 \pmod 3\}$, and let μ_K be the limiting distribution, as $x \to \infty$, of the primes $p \le x$ among the classes of $\mathcal{C}_K$. Then (SW1) holds, but $\mu_K(\mathcal{C}_K \setminus \mathcal{E}_K) = 1/2$ for every K and $\log|\mathcal{E}_K|/\log|\mathcal{C}_K| \to 1$. So (SW3) fails on both counts.*

Proof. $|\mathcal{C}_K| = \varphi(3^K) = 2\cdot 3^{K-1}$, which is exponential in K. By the prime number theorem for arithmetic progressions, μ_K is the uniform distribution on $\mathcal{C}_K$, and exactly half of the classes lie in $\mathcal{E}_K$. Finally $|\mathcal{E}_K| = 3^{K-1}$, so $\log|\mathcal{E}_K|/\log|\mathcal{C}_K| = (K-1)\log 3/(\log 2 + (K-1)\log 3) \to 1$. □

The first version asserted that "primes concentrate onto the inert sublattice $\mathcal{E}_K$" and that "the level of distribution 5/8 measures the depth of this concentration". The condition $p \equiv 2 \pmod 3$ constrains the smaller member of a twin-prime pair (and Sophie Germain primes $p > 3$), not primes in general, and Proposition 5.1 concerns all primes through $\Lambda(n)$. "Depth of concentration" was not defined, and 5/8 is not the exponent ratio of any set defined in this paper. Levels of distribution are downward closed (if a level θ holds, so does every smaller one), so 5/8 cannot be "the unique value consistent" with any structure. Condition W3 of [14] is true, with a one-line proof from the cyclicity of $(\mathbb{Z}/3^K\mathbb{Z})^*$, and does not bear on the level.

6. Structural Analogies

The first version called the parallels below "five structural correspondences" and labelled them proven; the five were never listed. We present them as informal analogies (matchings of roles), not as theorems, isomorphisms or functorial equivalences. The arithmetic case is withdrawn (Section 5); its column records what the first version proposed and what is actually the case.

6.1. **The analogy table.**

Role	Shannon AEP	Channel Coding	Cascade moduli (withdrawn)
Config. space	$\mathcal{A}^n$, size $\lvert\mathcal{A}\rvert^n$	$\mathcal{X}^n$, size $\lvert\mathcal{X}\rvert^n$	$(\mathbb{Z}/3^K\mathbb{Z})^*$, size $2{\cdot}3^{K-1}$
Constraint	i.i.d. law $p^{\otimes n}$	Channel $p(y\lvert x)$	Residue condition $a \equiv 2 \pmod 3$
Eff. subspace	Typical set $T_{\varepsilon_n}^{(n)}$	Largest reliable codebook	Classes $\mathcal{E}_K$; primes do not concentrate there
Gov. constant	$H/\log_2\lvert\mathcal{A}\rvert$	$C/\log_2\lvert\mathcal{X}\rvert$	None; 5/8 is not an exponent ratio here
Mechanism	Law of large numbers	Random coding; Fano	None established

6.2. **The formula analogy.** The first version asserted that all three governing constants obey

$$\theta^* = \frac{\text{total} - \text{constraint}}{\text{total}}.$$

Any $\theta \in [0,1]$ can be written in this form (take total T and constraint $(1-\theta)T$), so the form constrains nothing. Specifically:

- **AEP**: $H = \log_2\lvert\mathcal{A}\rvert - D(p\|u)$, where $D(p\|u)$ is the Kullback–Leibler divergence from the uniform distribution u on $\mathcal{A}$. This identity is correct, and gives $H/\log_2\lvert\mathcal{A}\rvert = 1 - D(p\|u)/\log_2\lvert\mathcal{A}\rvert$.
- **Channel Coding**: the first version wrote $C = \log\lvert\mathcal{X}\rvert - \min_{p(x)} H(X|Y)$. That formula is false: a point mass $p(x)$ gives $H(X|Y) = 0$, so it would give $C = \log\lvert\mathcal{X}\rvert$ for every channel. The correct formula is $C = \max_{p(x)} I(X;Y) = \max_{p(x)}\big[H(X) - H(X|Y)\big]$, a maximum of a difference rather than a fixed total minus a fixed constraint. For a symmetric channel it equals $\log_2\lvert\mathcal{Y}\rvert - H(\mathbf{r})$, where $\mathbf{r}$ is a row of the transition matrix [2, Ch. 7]; for the binary symmetric channel with crossover probability ϵ this is $1 - H(\epsilon)$.
- **Cascade moduli**: $\theta_W = (2^k - k)/2^k$ with $k = 3$, read as 8 configurations of which 3 are eliminated. The first version said that the Eisenstein filtration "provides $2^k = 8$ total configuration dimensions". The filtration has $2K$ steps and provides no such count, $\{0,1\}^3$ has 8 points (not 8 dimensions), and the elimination count is the postulate of [9, 13].

6.3. **The reduction analogy.**

- **AEP**: typical sequences are those whose empirical log-probability per symbol is close to the entropy.
- **Channel Coding**: in the random-coding proof of achievability, decoding uses joint typicality of codewords and channel outputs.
- **Cascade moduli**: no analogue. Primes are equidistributed between the inert and the split classes (Proposition 5.2).

7. Sophie Germain Primes

The first version presented Sophie Germain pairs $(p, 2p+1)$ as a "confirmed non-instance". It stated as a theorem, with a citation to itself, that they have dimensional parameter $k = 2$ because the residue of $2p+1$ modulo 3 is determined by that of p; that their predicted level of distribution is $\theta_2 = 1/2$; and that a computation gives $\delta \cdot \log x = 2.38 \pm 0.18$, "inconsistent with $\theta > 1/2$ at $p < 0.001$". All of this is withdrawn. We keep only the following remark.

Remark 7.1 (Sophie Germain pairs; heuristic). For $p > 3$, if $p \equiv 1 \pmod 3$ then $3 \mid 2p+1$, so Sophie Germain primes $p > 3$ satisfy $p \equiv 2 \pmod 3$. This is the same single condition as for the smaller member of a twin-prime pair, where the residue of $p+2$ is equally determined by that of p; the "coupling" argument for $k = 2$ would therefore apply to twin primes too. Since $\theta_1 = \theta_2 = 1/2$, assigning either depth predicts $1/2$, so the example cannot discriminate between depths and is not a falsification test. A "level of distribution of Sophie Germain pairs" is not a defined notion, and no such result is known. The quantity δ of the first version was never defined, and no finite computation can confirm or refute an asymptotic level of distribution.

The first version compared this coupling to a degraded channel, saying that a degraded channel has lower capacity "because extra dependence between input and output is introduced". That is backwards. Degradation $X \to Y \to Z$ destroys input–output dependence: $I(X;Z) \le I(X;Y)$ by the data-processing inequality, so the degraded channel has capacity at most that of the original. With this correction the analogy to "coupling reduces k" does not hold.

8. The k-Hierarchy

The formula $\theta_k = (2^k - k)/2^k$, obtained from the counting postulate of [9, 13], assigns a value to each depth k. Table 1 displays the first five levels with the status of each proposed instance.

Table 1. The k-hierarchy of constitutional sieve levels.

k	2^k	$2^k - k$	θ_k	Status
1	2	1	1/2	Bombieri–Vinogradov level 1/2 for primes [3, 8]; not derived from $(2-1)/2$
2	4	2	1/2	None established (Sophie Germain does not discriminate; Remark 7.1)
3	8	5	5/8	Cascade moduli: conjectural [12]; not an SW instance (Proposition 5.2)
4	16	12	3/4	No object specified (Remark 11.1)
5	32	27	27/32	No object specified

Remark 8.1. Note that $k = 1$ and $k = 2$ both give $\theta = 1/2$. Bombieri–Vinogradov gives level $1/2$ for primes on average over moduli $q \le x^{1/2}(\log x)^{-B}$; that $1/2$ comes from the large sieve inequality, not from a count of constraints. Levels beyond $1/2$ are known for primes with well-factorable weights and a fixed residue class: $4/7$ [4], $3/5$ [6], $66/107$ [5] and $5/8$ [7]. None of $4/7$, $3/5$ and $66/107$ is of the form $(2^k - k)/2^k$. The first version described $k = 2$ as "a

non-instance of the enhanced effect" and $k = 3$ as "the first genuinely new level"; no system of depth 2 or 3 is established.

9. The Constitutional Forcing Conjecture

Conjecture 9.1 (Constitutional Forcing Conjecture)**.** *Let $\mathcal{S}$ be a system in which:*

(i) the configuration space is exponential with growth rate $r > 0$;
(ii) the structural law is intrinsic (not an external parameter);
(iii) the system has depth k in the sense of the counting postulate of [9, 13]: exactly k of its 2^k binary configuration classes are eliminated by its k independent constitutional constraints.

Then $\mathcal{S}$ exhibits the Shannon–Wakil effect with governing constant $\theta^* = (2^k - k)/2^k$.

If a configuration $c \in \{0,1\}^k$ is called invalid when some constraint fails ($c_i = 0$ for some i), then k constraints invalidate $2^k - 1$ configurations, not k. Hypothesis (iii) is therefore a postulate about the system, not a consequence of its having k binary constraints.

Remark 9.2. The first version said that Conjecture 9.1 "encompasses information theory ($k = 1$: entropy/capacity)" and "prime constellations ($k = 3$: twin primes via Eisenstein structure)". Neither holds. The Shannon constants $H/\log_2|\mathcal{A}|$ and $C/\log_2|\mathcal{X}|$ are continuous functions of the source and the channel and take every value in $(0, 1)$, so they are not values θ_k, and in particular not $\theta_1 = 1/2$; and twin primes have depth 1 by the programme's own independence criterion (Section 5). No system is currently known that satisfies (iii) and exhibits the effect, so the conjecture is at present a proposal. It would be refuted by a system satisfying (i)–(iii) whose governing constant differs from θ_k.

10. Shannon–Wakil Effect vs. Analogies

It is important to distinguish the Shannon–Wakil effect from informal analogy.

An *analogy* between two fields observes superficial similarity and draws heuristic inspiration. The comparison is suggestive but not formally verifiable. Analogies are valuable but limited: they can be wrong in ways that are hard to detect, and they generate no falsifiable predictions.

The first version argued that the Shannon–Wakil effect differs from an analogy in three ways. As revised, the situation is as follows.

First, only the AEP satisfies Definition 2.1 (Proposition 3.2). Channel coding satisfies its size conditions only (Proposition 4.2), and the cascade case fails (SW3) (Proposition 5.2).

Second, the framework has not yet produced a falsifiable prediction. The Sophie Germain example cannot discriminate between depths (Remark 7.1), and no object is specified for $k = 4$ (Remark 11.1).

Third, the "formula correspondence" is not quantitatively precise: any $\theta \in [0, 1]$ can be written as (total − constraint)/total (Section 6.2).

The parallels of Section 6 are therefore, at present, informal analogies in the sense of the preceding paragraph.

11. Discussion and Open Problems

11.1. **Chronology.** Shannon's paper appeared in 1948 [1]. It contains the AEP (for ergodic Markov sources) and the channel coding theorem, the latter proved by a random-coding argument that was made rigorous later (Feinstein, 1954). The arithmetic case was proposed

in 2026 and is not established (Section 5). The first version argued that the gap between the two guarantees that neither result was constructed to match the other; since the arithmetic case is withdrawn, the chronology carries no evidential weight.

11.2. **No object for $k = 4$.**

Remark 11.1 (No object for $k = 4$). The formula gives $\theta_4 = (2^4 - 4)/2^4 = 12/16 = 3/4$. The first version predicted "a prime constellation with exactly $k = 4$ independent modular constraints, exhibiting level of distribution $\theta = 3/4$". No such object (sequence, family of moduli or sieve) is specified, and a level of distribution is a property of a sequence in arithmetic progressions, not of a constellation. So $\theta_4 = 3/4$ is not at present a testable prediction. Specifying such an object is the open problem.

11.3. **The ratio $5/4$.** The first version called $\theta_W/\theta_{BV} = (5/8)/(1/2) = 5/4$ the gain in level of distribution achieved by the Eisenstein structure over Bombieri–Vinogradov. That interpretation is withdrawn. Bombieri–Vinogradov is an average over all moduli $q \le x^{1/2}(\log x)^{-B}$, whereas the cascade family has $O(\log x)$ moduli. For that family the summed estimate holds trivially up to exponent 1 (Proposition 5.1), and the per-modulus level $5/8$ is conjectural. Levels above $1/2$ for general moduli are known with well-factorable weights (Section 8).

11.4. **Channel capacity and levels of distribution.** Both the channel capacity $C = \max_{p(x)} I(X;Y)$ and a level of distribution are defined as suprema. For the moduli 3^K the supremum of the admissible θ depends on the formulation: for the summed estimate of Proposition 5.1 it is 1, and in the per-modulus form it is unknown (Montgomery's conjecture predicts 1). If $5/8$ were proved for these moduli, it would be a lower bound for that supremum, not the supremum itself. The first version's claim that this "maximisation correspondence is the most precise of the five" is withdrawn.

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